Layer	Technology Components	Domain Expertise Needed
Agentic Workflow orchestrating tasks and human-machine collaboration	Cloud/Local Storage, Server Orchestration, Human-in-the-loop System, Cyber Security	**Organizational Insight** Deep understanding of organization's structure, workflow processes, and decision hierarchies
Agent-Tool Integration connecting LLMs to external power domain tools	MCP and API Integration, Streaming /Batch Data Integration, Error Recovery	**Tool and Software Expertise** Familiarity with power system engineering software (e.g., PSS®E, PowerWorld, PSCAD) and their functionalities
Foundation Models power-specialized knowledge and reasoning engine	RAG, Fine-tuned LLMs, Full Domain-Specific FMs, Benchmarking	**Power Sector Understanding** Knowing physical and economical basic of energy systems, regulatory, industry standards, and legal frameworks

Enable Scalability

Power Agent

Speed Up Deployment

PowerWF + PowerMCP + PowerFM

Qian Zhang, Le Xie. "PowerAgent: A Roadmap Towards Agentic Intelligence in Power Systems", IEEE Power and Energy Magazine, 2025